



Figure 1: Understanding a consensus mechanism

agree on one version of the truth based on shared data. This need has given rise to consensus processes as a primary protocol to allow distributed systems to come to agreement with each other.

Consensus mechanisms ensure that:

- All nodes validate transactions correctly.
- Only legitimate transactions are added to the blockchain.
- The network remains secure against malicious activities.
- Data consistency is maintained across all nodes.

The design of a consensus algorithm must balance three key aspects:

- **Security:** Protection against attacks and fraud
- **Scalability:** Ability to handle large volumes of transactions
- **Decentralisation:** Ensuring no single entity controls the network

Different blockchain platforms adopt different consensus models depending on their application requirements and performance constraints.

Byzantine Generals Problem

The Byzantine Generals Problem is a popular problem in computer science. It refers to the challenges of coming to an agreement when some group members are untrustworthy or trying to cause chaos, but must come to a consensus

about what they want to do. However, if they try to deceive others by giving them false or misleading information, it will be impossible for everyone to agree on a single course of action.

Obtaining proper functioning among several nodes (computers) in a blockchain system requires a method of compensating for incorrect or failed actions of those nodes in order to provide proper network operation. Systems that can withstand these types of node actions are called Byzantine fault tolerant systems.

The key challenges include:

- Detecting malicious nodes
- Preventing false transaction validation
- Maintaining agreement despite conflicting information

To address these challenges, modern blockchain systems implement consensus algorithms that are resilient to Byzantine faults, ensuring reliability and trust in decentralised environments.

Proof of Work (PoW)

Proof of Work (PoW) has been around since the early days of cryptocurrency and is the most popular method of achieving consensus in distributed networks today. Its prime directive is to maintain the security of the entire proof-of-work network by requiring the use of computer cycles as a resource to

validate a new transaction or create a new block.

Miners compete against one another to be the first to solve a complex mathematical problem. The problem is solving for an arbitrary number, known as the nonce, which when hashed with transaction information using a cryptographic hash function meets certain requirements.

The working process can be summarised as follows:

- Transactions are collected into a block.
- Miners attempt to solve a cryptographic puzzle.
- The first miner to find the correct nonce broadcasts the solution.
- Other nodes verify the solution.
- The block is added to the blockchain and the miner is rewarded.

This mechanism ensures that adding a block requires significant computational effort, making it difficult for attackers to manipulate the system.

The advantages of PoW are:

- Highly secure and resistant to tampering.
- Proven and reliable mechanism (used in Bitcoin).
- Strong validation through computational effort.
- Open participation (anyone can become a miner).
- Maintains integrity of the blockchain network.

However, there are a few disadvantages too:

- Very high energy consumption.
- Vulnerable to 51% attack if majority power is controlled.
- Mining power often becomes centralised in pools.
- Slow transaction processing speed.
- Limited scalability for large scale applications.

Proof of Stake (PoS)

A consensus mechanism that utilises probability, Proof of Stake (PoS) replaces computational work with economic participation to reach an agreement among